DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY



Report

Week 1 - Python, datasets and reproducible research

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Chapter 1

Work in this week

Create a project folder; open the supplied Week-1 seed-case JSONL; inspect 20 records with Python; verify IDs are unique, fields are present, and language/modality values are valid. Create a short README.

Chapter 2

Project Structure

2.1 Directory Tree

```
LLM_security/
|-- LICENSE
|-- configs/
|   `-- config.json
|-- datasets/
|   `-- raw/
|       `-- data_dictionary_v0.1.jsonl
|-- docs/
|   |-- README.md
|   `-- data_dictionary_v0.1.md
|
`-- project/
    |-- main/
    |   |-- __init__.py
    |   `-- validate_seed_dataset.py
    `-- src/
        `-- validate_seed.py
```

Listing 2.1: Repository layout (LLM_security)

2.2 Component Description

Path	Description
`config.json`	Central configuration file for the project
`data_dictionary_v0.1.jsonl`	Dataset version v0.1, more details in data dictionary.
`docs/README.md`	Main readme for the entire project, more details in GitHub repository.

Path	Description
`validate_seed_dataset.py`	Main run validate the seed dataset of data_dictionary_v0.1
`validate_seed.py`	Function of validating the seed dataset

Chapter 3

Validate seed-case report

1. Check whether any fields contain null values or values that fall outside the defined categories.
2. Verify whether the content of the prompt accurately aligns with the defined categories.

Limitation

- Some fields are self-defined: neutral communitation style and ambiguity level, both fields are semantic judgments